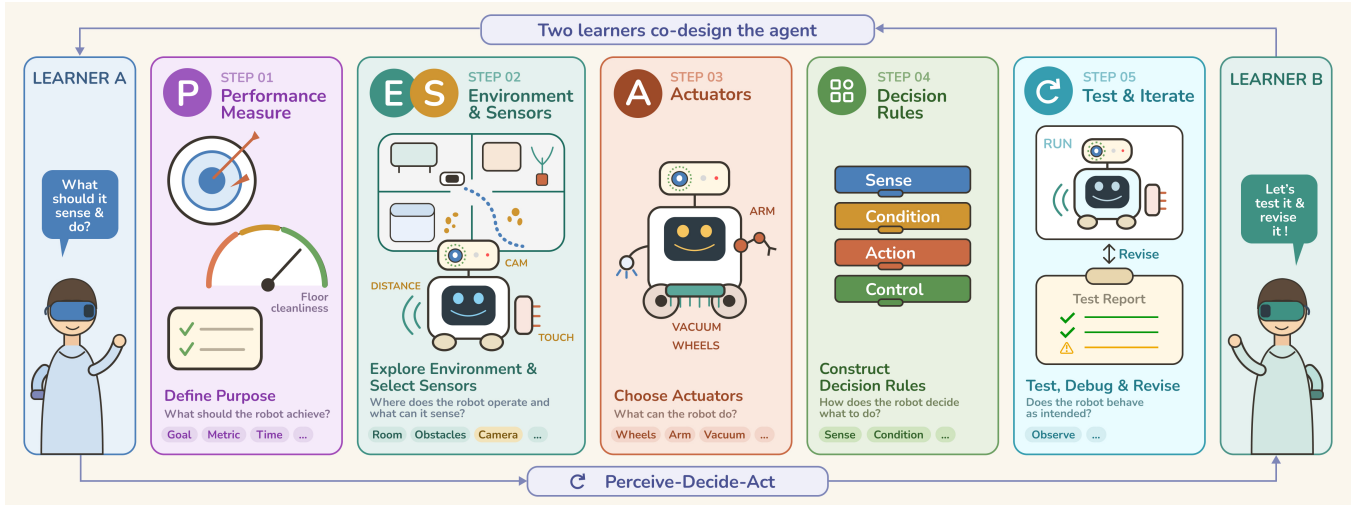


# CollabPEAS: A Collaborative VR System for Learning AI Agent Design Through the PEAS Framework

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**Figure 1: Overview of CollabPEAS.** Two learners collaboratively specify a virtual cleaning robot in a shared VR environment, working through a five-step flow that operationalizes the PEAS framework. The first four steps define a performance measure, characterize the environment, select sensors, and choose actuators, making each specification choice a spatially grounded and jointly inspectable decision. In the final Build and Iterate step, learners assemble SENSE → CONDITION → ACTION → CONTROL decision rules and then test, debug, and revise the agent. Test results feed back to refine the earlier PEAS choices, framing agent design as an iterative, collaborative activity in which every decision connects to observable robot behavior.

## Abstract

We present CollabPEAS, a collaborative system for artificial intelligence (AI) literacy in which pairs of learners come to understand how an AI agent's behavior is produced by human specification. Working in shared virtual reality (VR), learners jointly specify a virtual cleaning robot using the PEAS framework (Performance measure, Environment, Actuators, Sensors), developing competencies for reasoning about how AI agents sense, decide, and act. The system integrates three coordinated design elements: (1) a five-step PEAS-guided design flow, implemented in VR, that makes agent specification choices spatially grounded and observable, (2) a block-based coding interface that preserves the perceive, decide, act reasoning cycle, and (3) collaboration supports designed to structure negotiation and shared ownership between partners. We

situate CollabPEAS within AI literacy and collaborative learning research and discuss directions for future evaluation.

## CCS Concepts

• **Social and professional topics** → **Computing education**; • **Human-centered computing** → **Collaborative and social computing systems and tools**; **Virtual reality**; *Computer supported cooperative work*; • **Applied computing** → **Interactive learning environments**; *Collaborative learning*.

## Keywords

AI literacy; Collaborative learning; PEAS framework; Virtual reality; Block-based coding; Agent specification; Computer-supported cooperative work

## ACM Reference Format:

Wanru (Jovie) Li, Yichen (Andy) Yu, and Qiao (Georgie) Jin. 2026. CollabPEAS: A Collaborative VR System for Learning AI Agent Design Through the PEAS Framework. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '26)*, October 10–14, 2026, Salt Lake City, UT, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3785651.3831526>



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© 2026 Copyright held by the owner/author(s).  
ACM ISBN 979-8-4007-2378-0/2026/10  
<https://doi.org/10.1145/3785651.3831526>

## 1 Introduction

Artificial Intelligence (AI) agents perceive their environment through sensors and act through actuators in pursuit of goals. They are now embedded in everyday systems [16], from voice assistants and recommendation systems to autonomous navigation [8]. Although people may interact with these systems regularly, the design decisions behind the agent often remain hidden. This gap is central to AI literacy education, where learning AI requires more than learning how to use AI tools. Learners also need to understand how agents sense, decide, act, and how their behavior is shaped by human-authored specifications [11]. However, for most learners, the process of specifying an AI agent remains opaque.

The PEAS framework offers a way to make this hidden specification process more explicit. PEAS structures agent design around four interdependent components: Performance measure, Environment, Actuators, and Sensors [16]. By asking learners to define what an agent should achieve, where it operates, how it can act, and what information it can sense, PEAS turns the behavior of the agent into a set of design choices that can be examined and discussed. In doing so, it provides a useful scaffold for translating professional agent design reasoning into learning-oriented activities.

Using this scaffold in a learning activity also requires attention to how learners engage with AI concepts. Research on AI literacy education points to the value of collaborative learning, as peer explanation can help learners remain engaged with abstract concepts [11]. This focus on collaboration aligns with agent specification, which often requires people to negotiate task definitions, weigh sensor and actuator trade-offs, and interpret emergent behavior [16]. These considerations motivate our system, which frames the specification of PEAS-based agents as a collaborative activity. Immersive Virtual Reality (VR) is well-suited for this activity, as it supports spatial co-presence, allows partners to jointly observe and discuss agent behavior [6], and grounds abstract PEAS components as visible, manipulable objects [7, 10], making sensing, acting, and environmental constraints directly observable in ways that are difficult to support through text-based or 2D interfaces [11].

We present CollabPEAS, a collaborative VR system for AI literacy education in which learner dyads specify a virtual cleaning robot using the PEAS framework and block-based coding, coming to understand how agent behavior is produced by human specification. The system integrates three design elements: a VR environment that makes PEAS components spatially grounded and observable, a block-based coding interface that preserves agent design reasoning while reducing the burden of programming syntax, and collaboration supports designed to scaffold productive dyadic specification work.

## 2 Related Work

### 2.1 AI Literacy and the PEAS Framework

AI literacy refers to the competencies that enable people to critically evaluate AI technologies, collaborate with AI, and use AI in daily life and work. These competencies are essential for informed participation in an AI-mediated world. Long and Magerko [11] provide a comprehensive framework that identifies seventeen competencies and fifteen design considerations; CollabPEAS specifically targets five of these: Recognizing AI (C1), Decision-Making (C8),

Action and Reaction (C14), Sensors (C15), and Programmability (C17). Existing AI literacy interventions for non-technical learners have used media-rich curricula, hands-on machine learning activities, and block-based environments such as Scratch [14] and related tools [18], helping learners understand what AI is and how it operates at a conceptual level.

However, a remaining challenge is the lack of designs that let learners practice the structured specification skills used to design AI agents: existing approaches primarily teach *about* AI agents, with fewer opportunities to reason through what an agent should sense, how it should act, and how success should be evaluated. In response, we use PEAS as a guiding scaffold for learner-facing agent design. PEAS [16] decomposes agent design into four components: Performance measure, Environment, Actuators, and Sensors. Although PEAS is already used as a teaching scaffold in introductory AI education, our contribution is to adapt it into an embodied collaborative experience that makes each component a target for AI-literacy competency development.

### 2.2 Collaborative Learning in Immersive VR

Collaborative learning can effectively support AI literacy competencies. As Long and Magerko [11] note, collaborative activities can sustain engagement and promote conceptual understanding through peer explanation. Previous work has explored different collaboration strategies, such as role-based collaboration [17] and flexible adaptive collaboration modes [20], to support AI literacy and related computational competencies such as programming.

In this work, we use immersive virtual reality (VR) to support collaborative learning of AI literacy in AI agent design. VR provides a shared spatial environment where learners can observe agent behaviors, discuss design choices, and connect AI concepts with embodied interaction. Previous research on collaborative VR found that VR can improve knowledge acquisition, social presence, and collaboration quality [6]. Interactive VR environments can increase the germane cognitive load when embodied interactions are conceptually meaningful rather than decorative [5]. For AI agent specification, VR is especially suitable because PEAS components are spatial and physical: sensors perceive, actuators act, and environments constrain. VR makes these components observable and jointly inspectable, transforming the specification into an embodied social activity where partners can see, discuss, and manipulate shared agent components [6, 7, 10]. This aligns with the theory of embodied interaction, which argues that spatial engagement with a system can shape how learners reason about it [3].

## 3 Design Goals

CollabPEAS is designed to help non-technical learners understand AI agents as systems whose behavior results from connected choices about performance measures, environment, sensors, actuators, and decision rules. The three design goals below (DG1–DG3) state what learners must achieve; Sections 4.1 and 4.2 describe how each is performed.

- **Make the PEAS framework observable and consequential(DG1).** Existing AI literacy interventions often teach *about* AI agents conceptually, but rarely let learners see how specification decisions produce agent behaviors [11].

Because PEAS links performance measures, environments, actuators, and sensors with resulting behavior [16], learners need to see the consequences of their choices, not only their labels. We therefore set DG1: every specification decision must be immediately observable in the robot’s behavior.

- **Support learning through progressive scaffolding(DG2).** PEAS involves four interdependent components that must be understood together, but presenting them all at once can overwhelm non-technical learners [11]. Structured progressions in block-based environments support computational thinking without programming prerequisites [14, 19]. For example, learners should grasp why a performance measure matters before choosing sensors to achieve it, not encountering all four components at once. We therefore set DG2: learners must build their specification one component at a time, preserving how each constrains the next.
- **Collaborative scaffold agent design(DG3).** The specification of AI agents is collaborative in practice [16], and collaboration matters for the learning of AI literacy [11]. However, unstructured collaboration can be less productive for novices [2, 17], while fixed roles can limit shared ownership [20]. For example, if either partner could unilaterally finalize a sensor choice, one learner could dominate while the other disengages. We therefore set DG3: partners must negotiate and jointly own each decision, not work in parallel or under fixed roles.

## 4 System Design

CollabPEAS is a multiplayer VR learning environment built in Unity 6 for Meta Quest headsets, using Photon Fusion 2 with Shared Authority topology for real-time synchronization. Both participants occupy the same virtual space with access to all shared objects. The system has two main components: the PEAS-Guided Design Flow (Section 4.1), which addresses DG1 and DG2, and the Collaboration Design Supports (Section 4.2), which address DG3.

### 4.1 PEAS-Guided Design Flow

The learning experience to realize DG1 (observability) and DG2 (progressive scaffolding) is a five-step design flow in which dyads collaboratively specify a household cleaning robot, tying each PEAS choice to observable behavior (DG1) while introducing components incrementally rather than all at once (DG2).

- **Define purpose and performance measures.** Participants identify what their robot is for and how to evaluate its success, distinguishing a goal (“clean the apartment”) from a performance measure (“percent of floor area cleared in 10 minutes”) that determines how the agent behaves.
- **Characterize environment and select sensors.** Participants walk through the 3D apartment and select which sensors (camera, distance, touch, etc.) to mount on their robot; different sensor choices yield different robot behaviors in the same environment, making perception and capability directly observable (DG1).
- **Choose actuators.** Participants select the physical capabilities (wheels, arms, cleaning tools) that define what the robot can do, connecting each selection back to the performance

measure: if the measure requires a behavior, the actuators must be capable of producing it.

- **Program agent behavior through block-based coding.** Participants specify the robot’s decision logic by arranging pre-coded draggable 3D block cards into four-slot chains: SENSE → CONDITION → ACTION → CONTROL. The system enforces compatibility between blocks: a Distance Sensor block returns NEAR, FAR, or BLOCKED, so only an obstacle condition such as IF obstacle NEAR can snap into Slot 2, while a condition reading a different sensor’s value, such as IF floor DIRTY, bounces back with a brief label. Each completed chain is a discrete decision rule; because all rules run in parallel and whichever sensor fires first on a given tick triggers its rule’s action, the robot’s behavior emerges from the environment rather than a fixed script. No programming knowledge is required [9], which operationalizes DG2 by letting non-technical learners engage directly in decision-rule construction.
- **Test, debug and revision;** Participants observe their robot running in the apartment and identify mismatches between the intended and observed behavior. The debug loop is scaffolded as design reasoning rather than error correction: the productive question is not “what is broken?” but “is the agent rational given what we have specified, and if not, which PEAS component needs to change?”

### 4.2 Collaboration Design Supports

To address DG3 (scaffolded collaboration), the system offers five collaboration supports for how partners coordinate within the PEAS workflow, communicating throughout via in-VR voice chat. The supports occupy a middle ground between fully symmetric access and fully role-assigned collaboration, drawing on evidence that unstructured collaboration can be less effective than structured alternatives [17] and that shared control with awareness tools improves collaboration quality in VR [6].

- **Agree-before-placing.** When one participant drags a block toward a placement slot, it enters a pending state visible to both; the second participant must approve, request discussion, or suggest an alternative before the placement takes effect. The mechanism keeps every specification decision explicitly agreed on, creating a window for verbal negotiation.
- **Alternating lead-reviewer roles.** Each design step assigns one participant as lead (initiating placements) and the other as reviewer (approving, rejecting, or suggesting changes), with roles alternating each step so that both experience both perspectives. This balances evidence that role-based structure improves learning outcomes [17] with preserving shared ownership, since alternating assignments avoid permanently excluding either participant.
- **Structured inter-step reflection prompts.** At each transition between steps, the system pauses and presents a joint reflection question both participants must answer before proceeding, e.g., after sensor selection: “Do you both agree these sensors give the robot enough information to do its job?” These prompts aim to create checkpoints for alignment and connect each new decision back to those already made.

- **Contribution awareness display.** A persistent visual indicator will show each participant's relative contribution (blocks placed, log entries, debug moves), updated in real time. Rather than scoring performance, it functions as a mutual awareness tool for self-regulating contribution equity, drawing on evidence that shared awareness tools improve communication quality in collaborative VR [6].
- **Dual observation logging.** During Test, debug, and revise, both participants will independently mark observations in a shared log as they watch the robot execute; the system then highlights agreement or divergence, creating discussion points for debugging. The goal is to turn debugging from a single-perspective activity into a collaborative comparison of interpretations [12].

These supports aim to structure the joint specification enough to prevent coordination failures associated with fully symmetric access [17], while preserving the shared ownership that defines collaborative problem solving [15]. Such a structure can also introduce friction, slow exploration, or add pressure to negotiate every move; we treat the right level of structure as an open question our planned evaluation will examine, including whether some supports should be loosened as partners become more familiar with the task.

## 5 Conclusion and Future Work

CollabPEAS is a collaborative AI-literacy system currently under active development. The PEAS-guided design flow, the block-based coding interface, and the VR environment are implemented in Unity 6 with Photon Fusion 2 networking, and the five collaboration supports are fully specified.

Once these supports are implemented, we plan to evaluate the complete system through a mixed-methods between-subject study with non-technical participants recruited on campus, testing in August, comparing the VR condition against a functionally equivalent 2D desktop interface. This comparison is designed in part to assess whether the learning advantages persist beyond the novelty effects documented in VR learning research [4]. The evaluation will assess the AI literacy outcomes through pre/post knowledge assessments aligned with the targeted competencies of [11], attitudinal changes through an adapted AI Literacy Questionnaire [13], and collaborative dynamics through behavioral logs, mid-session collaboration checks and semi-structured interviews analyzed through thematic analysis [1].

Within this evaluation, a central focus is whether the collaboration supports produce more equitable participation and more verbal negotiation than would emerge from fully symmetric access alone, and whether this, in turn, is associated with a deeper conceptual understanding of how the PEAS components relate to agent behavior [17, 20]. Whether this expectation holds, and how the five supports compare to one another remain open empirical questions. We hope that the findings will be useful to researchers and designers in the Computer-Supported Cooperative Work (CSCW) community who work on collaborative tools for technical learning.

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